Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

Solutions

1.

a) State two advantages and two disadvantages of encoding analogue signals in time rather than amplitude.

Advantages:

- Supply independence
- Spike domain coding can be event driven resulting in low power (triggers processing) [2 points]

Disadvantages

- Specialised circuits are required for encoding.
- Data is quantised in time, and therefore susceptible to phase noise. [2 points]

[4]

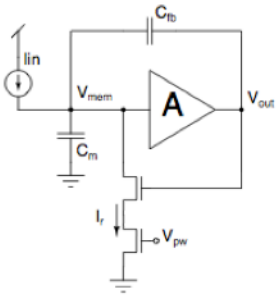
b)

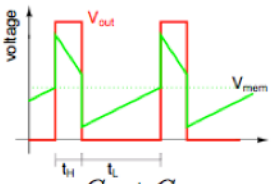
i. Draw a circuit that encodes a current mode signal in a domain that is continuous in time but discrete in amplitude.

[4]

ii. Explain its operation using both equations of the relationship between the input and output signals as well as time domain plots of the amplitude of an encoded analogue signal.

Axon Hillock circuit:
Circuit Diagram and explanation of operation (4 points)
Equations (2 points)
Time domain plot (2points)





$$t_L = \frac{C_{fb} + C_m}{I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_{in}} V_{dd}$$

$$t_H = \frac{C_{fb} + C_m}{I_r - I_{in}} \Delta V_{mem} = \frac{C_{fb}}{I_r - I_{in}} V_{dd}$$

Frequency $\propto I_{in}$

Pulse width $\propto 1/I_r$ for $I_r \gg I_{in}$

[4]

c) For the cascade of amplifiers shown in Figure 1.1, derive the output referred noise and explain what factors can be optimized to improve the signal to noise ratio.

[8]

Mos Noise: $$v_n^2 = \frac{K_w}{I_D^p} \cdot \Delta f + \frac{K_f}{A} \cdot \ln\left(\frac{f_h}{f_l}\right)$$

Distributed gain: $$G_i = \prod_{k=i}^{k=M} g_k$$

Solutions

Total output Noise: $$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2$$

Final Equation: $$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2 = \sum_{i=1}^{i=M} \left(n_i \frac{K_w}{(I_i / n_i)^p} \cdot \Delta f + n_i \frac{K_f}{(A_i / n_i)} \cdot \ln\left(\frac{f_h}{f_l}\right) \right) G_i^2$$ [6 points]

- Increasing Power and Area reduce the noise
- If we are willing to compute slowly (Δf and fh/fl are both small) then we can keep output noise low without consuming a lot of power. [2 points]

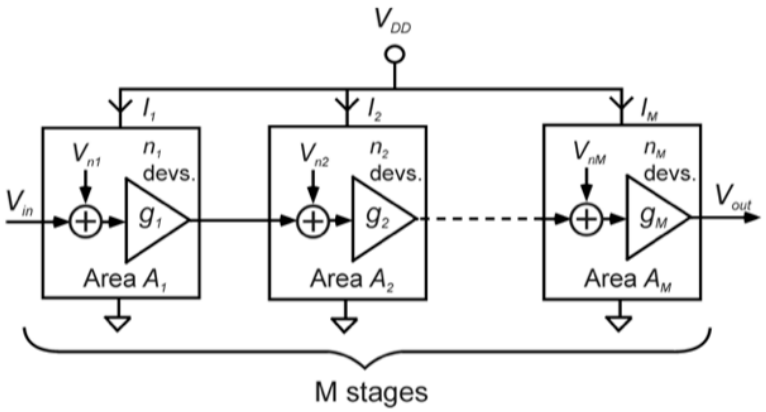


Figure 1.1: Cascade of amplifiers

Solutions

2. This question concerns translinear Circuits.

a) Give two limitations when using MOSFETs in translinear circuits.

Drawbacks:
Currents need to be limited to below 100nA. [1]
Temperatures must be equal to get rid of thermal voltage terms. [1]

[2]

b) Figure 2.1 shows a translinear circuit.

i. Derive the function of the circuit using the translinear principle. I_z is the output and I_x is the input.

Squaring Circuit from notes:

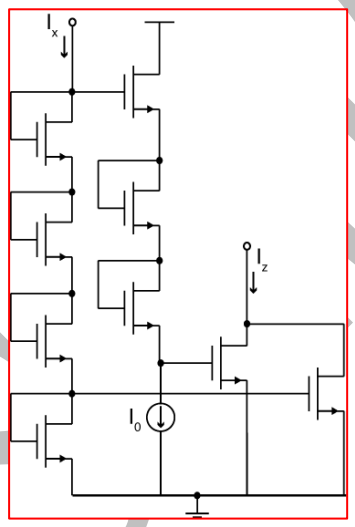
$I_1 . I_2 = I_3 . I_4$ and therefore $I_Z = \frac{(I_X)^2}{I_0}$ Total 2 points.

[2]

ii. Show how, with the addition of a few transistors, the circuit can be modified to calculate the function:

$$I_Z = \frac{(I_X)^4}{I_0^3} - I_x$$

New circuit derivation: $I_Z = \frac{(I_X)^4}{I_0^3} - I_x$



Total 5 points.

[4]

iii. A capacitor is added between the point A on the circuit in Figure 2.1 and ground. Derive the new transfer function of the circuit stating the gain and time constant. The final equation of the transfer function should be represented in the s-domain.

Derivation from theory:
Capacitor current:

Solutions

$$I_c = CnU_t \cdot \frac{\dot{I}_Z}{I_Z}$$

$$I_Z = \frac{(I_X)^2}{I_0 + I_c}$$

Combining:

$$I_z = \frac{I_x^2}{I_o + CnU_t \cdot \frac{\dot{I}_z}{I_z}}$$

$$I_z I_o + CnU_t \cdot \dot{I}_z = I_x^2$$

Taking laplace transform:

$$I_z I_o + CnU_t \cdot sI_z = I_x^2$$

$$I_z(I_o + CnU_t \cdot s) = I_x^2$$

$$I_z = \frac{I_x^2}{I_o + CnU_t \cdot s} = \frac{I_x^2 / I_0}{1 + CnU_t \cdot s / I_0}$$

Gain = I_x^2 / I_0

Time constant: $\tau = CnU_t / I_0$

[4]

c) Figure 2.2 shows a translinear circuit capable of computing the trigonometric function sin(πy) through the approximation $\frac{y-y^3}{1+y^2}$.

i. Express the currents I_3 and I_4 in terms of I and I_X.

[4]

New Derivation

$$\left[\frac{I - I_X}{2}\right]^2 II_4 = I_3 \left[\frac{I + I_X}{2}\right]^2$$

$\frac{I_4}{I_3} = \left[\frac{I+I_X}{I-I_X}\right]^2$ and $I_3+I_4=I$

$$I_4 = \frac{I(I + I_X)^2}{(I - I_X)^2 + (I + I_X)^2}$$

$$I_3 = \frac{I(I-I_X)^2}{(I-I_X)^2+(I+I_X)^2}$$

[4]

ii. Express the differntial output current I_2-I_1 and show that when $I_X = yI$ then I_2-$I_1 = I\frac{y-y^3}{1+y^2}$

[4]

$$I_2 = \frac{I - I_X}{2} + \frac{I(I + I_X)^2}{(I - I_X)^2 + (I + I_X)^2}$$

$$I_1 = \frac{I + I_X}{2} + \frac{I(I - I_X)^2}{(I - I_X)^2 + (I + I_X)^2}$$

$$I_2 - I_1 = \frac{I^2 I_X - I_X^3}{I^2 + I_X^2}$$

$$I_2 - I_1 = I\frac{y - y^3}{1 + y^2}$$

Solutions

[4]
[Total 8 points]

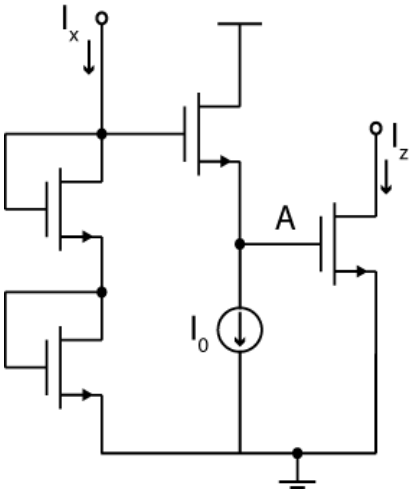


Figure 2.1

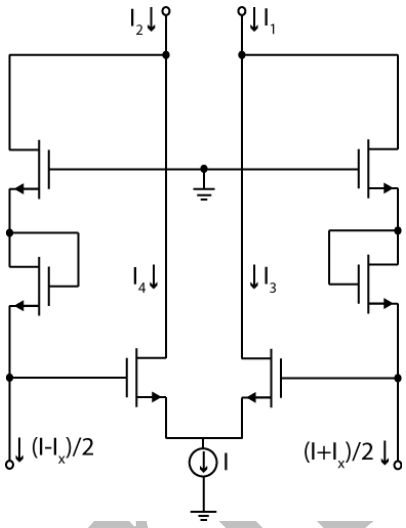


Figure 2.2

Solutions

3.

a) Explain the principle of companding, illustrating how the dynamic range changes through the signal processing chain.

[Bookwork]
Definition [2]
Diagram [2]

[4]

b) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

New derivation:

$$Y = I_{out} = I_0 \sinh(V_c / nU_t) = g(V_C)$$

$$\frac{d(g(V_c)}{dV_C} = \frac{dY}{dV_C} = \frac{dI_{out}}{dV_C} = \frac{I_0}{nU_t}\cosh(V_C / nU_t) = \frac{I_{Cosh}}{nU_t}$$

$$f(x) = I_C = X\tau C\left(\frac{dY}{dV_C}\right)^{-1} = X\tau C\frac{nU_t}{I_{\cosh}} = X\frac{I_1}{I_{\cosh}} \qquad X = I_{in}, \quad I_1 = \tau \cdot CnU_t$$

$$I_C = f(I_{in}) = \frac{I_{in}I_1}{I_{\cosh}}$$

[6]

c) The transfer function of a biquad filter is defined in state space representation by the following equations:

$$\dot{X}_1 = \frac{-\omega_0}{Q}X_1 - \omega_0 X_2 + \omega_0 U_1$$

$$\dot{X}_2 = \omega_0 X_1 - \omega_0 U_2$$

$$Y = X_1$$

whereby Y is the output and U_1 and U_2 are the inputs and X_1 and X_2 are the states.

i. By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

Solutions

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U_2 = I_{U2} \exp\left(\frac{V_{U2}}{nU_t}\right)$$

New derivation from Theroy:

$$\frac{I_1}{nU_t}\exp\left(\frac{V_1}{nU_t}\right)\dot{V}_1 = -\frac{\omega_0}{Q} I_1 \exp\left(\frac{V_1}{nU_t}\right) + \omega_0 I_2 \exp\left(\frac{V_2}{nU_t}\right)$$

$$\frac{I_2}{nU_t}\exp\left(\frac{V_2}{nU_t}\right)\dot{V}_2 = -\omega_0 I_1 \exp\left(\frac{V_1}{nU_t}\right) + \omega_0 I_U \exp\left(\frac{V_U}{nU_t}\right)$$

$$C_1\dot{V}_1 = -\frac{C_1\omega_0 nU_t}{Q} + C_1\omega_0 nU_t \frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C_2\dot{V}_2 = -C_2\omega_0 nU_t \frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right) + C_2\omega_0 nU_t \frac{I_U}{I_2}\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

Let C1=C2=C and define $I_\omega = C\omega_0 nU_t$

$$C\dot{V}_1 = -\frac{I_\omega}{Q} + I_\omega \frac{I_2}{I_1}\exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C\dot{V}_2 = -I_\omega \frac{I_1}{I_2}\exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_\omega \frac{I_U}{I_2}\exp\left(\frac{V_U - V_2}{nU_t}\right)$$

A suitable mapping:

$$I_1 = I_2 = I_\omega$$

$$I_U = I_0$$

Final equations:

$$C\dot{V}_1 = -\frac{I_\omega}{Q} + I_\omega \exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

$$C\dot{V}_2 = -I_\omega \exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_0 \exp\left(\frac{V_U - V_2}{nU_t}\right)$$

$$Y_1 = I_\omega \exp\left(\frac{V_1}{nU_t}\right)$$

$$Y_2 = I_\omega \exp\left(\frac{V_2}{nU_t}\right)$$

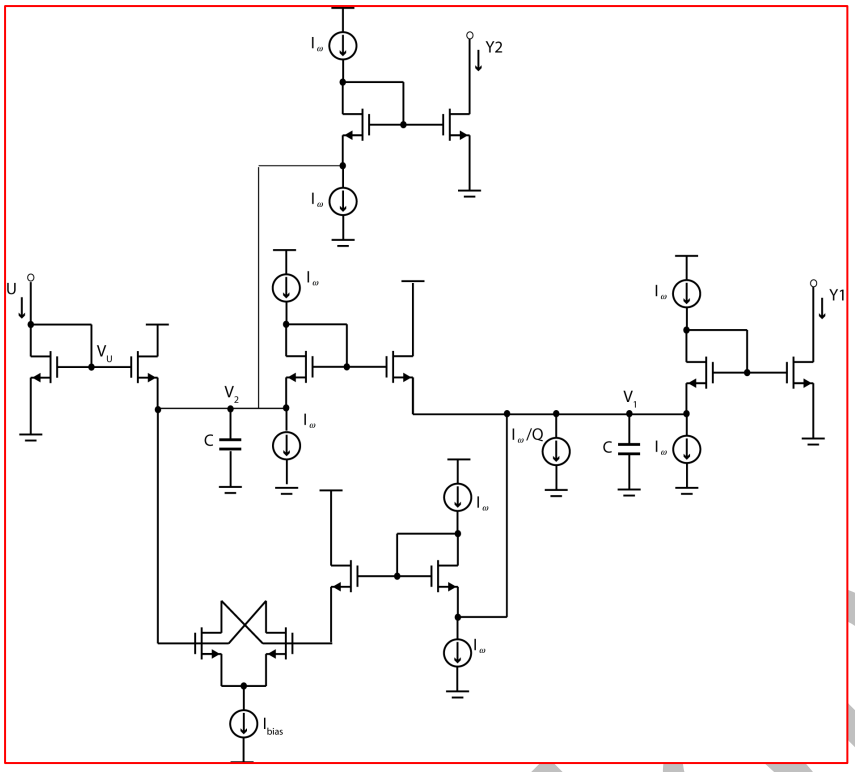
[5]

ii. With these log-domain design equations, sketch a schematic of the final log domain filter using weak inversion MOS transistors.

[5]

A Solution using the above equations:

Solutions


Y2
U
Y1
C
C
Solutions

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

i. Explain its operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[bookwork]

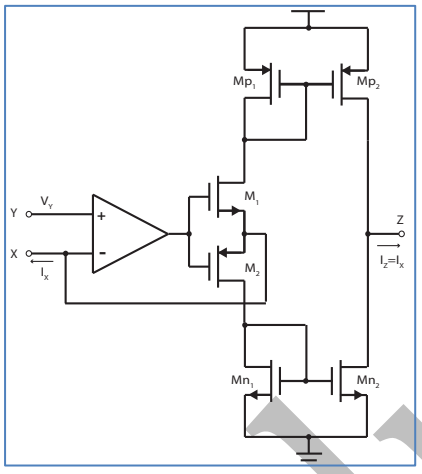
- Current-voltage characteristics
  $V_X=V_Y$, $I_Z=\pm I_X$, $I_Y=0$
  $Z_Y \rightarrow \infty$, $Z_X \rightarrow 0$, $Z_Z \rightarrow \infty$ [2]
- Voltage-follower between Y-input and X-output
- Current-follower between X-input and Z-output [1]

Plus a comparison with Op-amp terminals [1]

[4]

ii. Draw the circuit of a CMOS implementation of a bi-directional CCII+ with high output impedance.

Schematic:



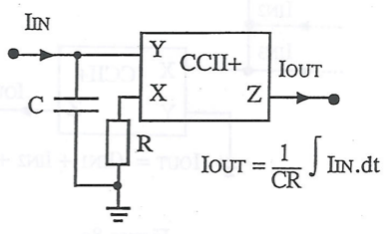
[3]

We can use cascaded current mirrors to increase output impedance at the expense of dynamic range. [1]

[4]

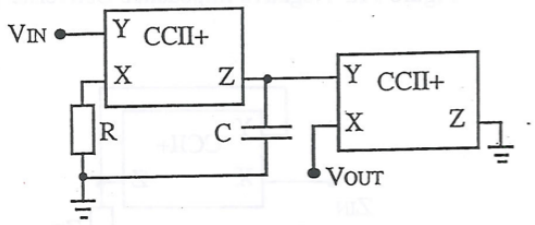
iii. Using the current conveyor in Figure 4.1, draw the schematics of a current mode integrator and a voltage mode integrator, each with a time constant, τ =1s. You may assume R = 100KΩ.

Current integrator [2]:



Voltage integrator [2]:

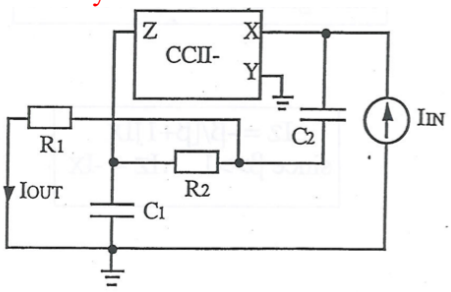
Solutions



[4]

b) Figure 4.2 shows a voltage mode Sallen key filter. Using the adjoint principle transform this to it's current mode equivalent using CCII+ blocks.

Sallen Key Circuit with CCII-:



[2]

Student also needs to realize the CCII- block required can be implemented using two CCII+ blocks. [2]

[4]

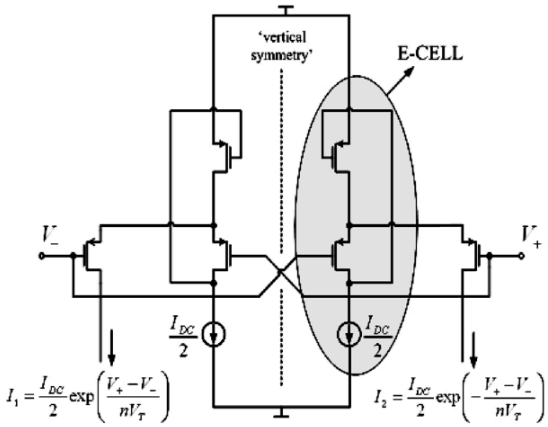
c) Sketch a current mode circuit that can implement the following hyperbolic functions:

$$I_1 - I_2 = I_{DC}\sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_1 + I_2 = I_{DC}\cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

[4]

[bookwork]



$$I_1 - I_2 = I_{DC}\sinh\left(\frac{V_1 - V_2}{nU_t}\right)$$

$$I_1 + I_2 = I_{DC}\cosh\left(\frac{V_1 - V_2}{nU_t}\right)$$

Solutions

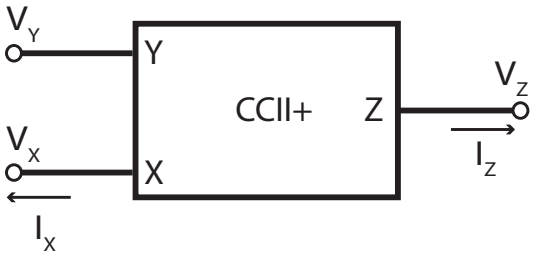


Figure 4.1

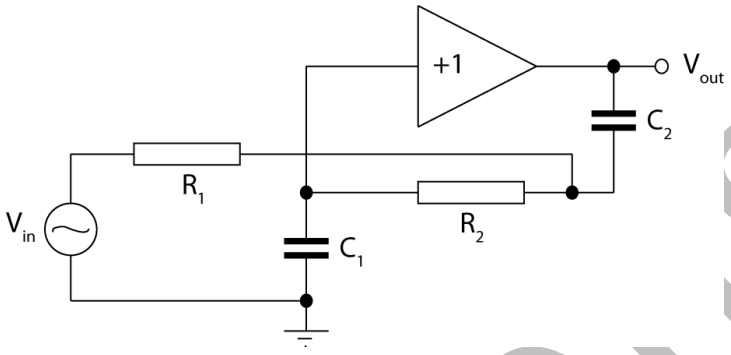


Figure 4.2

Solutions

5. This question relates to discreet time integrators.

a) Figure 5.1 shows a discreet time RC integrator.

i) Derive it's time constant of integration.

Switches M1, M2 and C1 for a resistor R, and time constant $\tau = RC_2$ where

$$R = \frac{1}{C_1 f_{CK}}$$

[2]

ii) Assuming C_1=C_2, select a suitable clock frequency to give a time constant τ =1µs

[2]

$\tau = C_2 / C_1 * f_{ck}$ which results in f_{ck}=1MHz

iii) For the circuit Figure 5.1, when CK goes low all of the channel charge, Q_1, of transistor M_1 is injected into C_1. This causes an error voltage ΔV_{C1} in C_1 defined by:

$$\Delta V_{C1} = \frac{Q_1}{C_1}$$

Then transistor M_2 turns ON. The channel charge of M_2 is split equally between C_1 and C_2. Estimate the total amount of charge injection and therefore error voltage ΔV_{C2} from switches M_1 and M_2 which appears on the output capacitor C_2 after CK goes high again, that is after one integration cycle. [4]

[New application of taught theory]

Charge in Q_1 is injected into C_1 when CK goes low and M_1 turns off.

M_2 Turns on so charge $Q_1/2$ appears between both capacitors C_1 and C_2.

Charge is absorbed by M_2 to create channel but is injected back equally between C_1 and C_2 when CK goes high again. So the remaining charge injection on C_2 is $Q_1/2$.

Therefore $\Delta V_{C2} = \frac{\Delta V_{C1}}{2}$ [4]

b) Figure 5.2 shows an autozeroing switched capacitor integrator capable of removing any offset in the amplifier.Derive the output voltage of the integrator during phase Φ1 and Φ2 showing how the offset is removed showing the charge on each capacitor.

[Derivation from notes]

Student should conclude to the following equations showing at each phase how the offset V_{os} gets subtracted.

$$v_{out}^{\Phi 2}(n) = v_{out}^{\Phi 2}(n-1) - (C_1 / C_2) v_{in}^{\Phi 2}(n)$$

$$v_{out}^{\Phi 1}(n) = v_{out}^{\Phi 2}(n) + v_{os}$$

[5]

c) Figure 5.2 shows a schematic of switched current integrator.

i) Derive the transfer function of the switched current integrator and show that in the z-domain this may be represented as H(z)= A/(z-1).[5]

[derivation from notes]

ii) Explain one limitation of the switched current technique that doesn't exist in its voltage mode equivalent.

[2]

VDS variations of memory transistors are an issue causing mismatch, need to be compensated [2].

Solutions

6. Figure 6.1 shows a biomedical system used to measure the similarity between two EEG channels by recording action potentials from the human brain. The front end consists of a differential operational amplifier (OA) connected to a voltage-to-current (V/I) converter. This then connects to a current rectifier followed by a peak detector that detects the envelope of the converted current I_{out1}. The similarity between the envelopes of the two channels, I_{env1} and I_{env2}, is detected using a current mode correlator to derive an output signal I_{corr}.

a) Explain one method to make the input signals V_{sig1} and V_{sig2} immune to 1/f noise in the operational amplifier, and draw a block diagram of this method illustrating how the frequency spectrum of the signal changes during each stage.

[5]

[bookwork]
Student needs to identify that chopping can be used and explain the principle of chopper stabilization with the following diagrams.

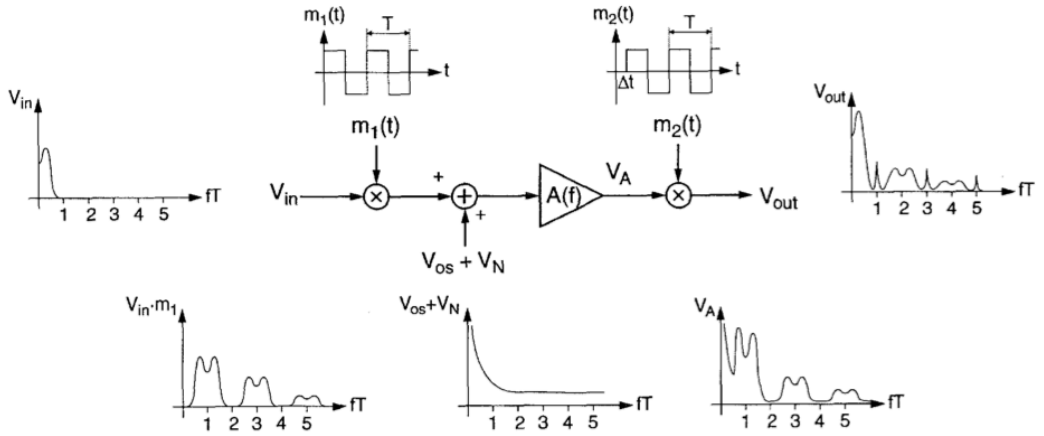
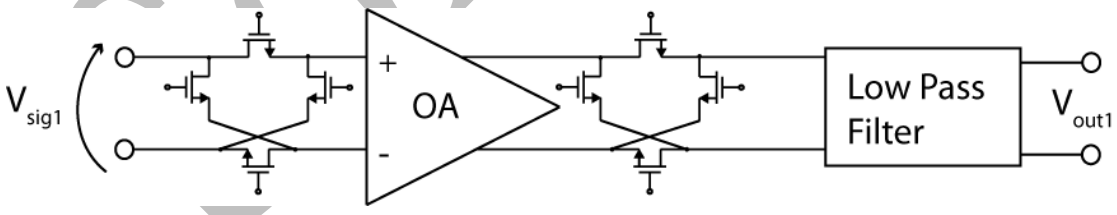


Fig. 9. The chopper amplification principle [19].

b) Design a suitable front end for this system using the method described in part 6.a. You may use the operational amplifier as in Figure 6.

[5]

[Application of theory].
Student needs to add chopping switches in input and output and low pass filter block.



c) Figure 6.2 show a suitable circuit that can be used as a current mode peak detector. Explain it's principal of operation.

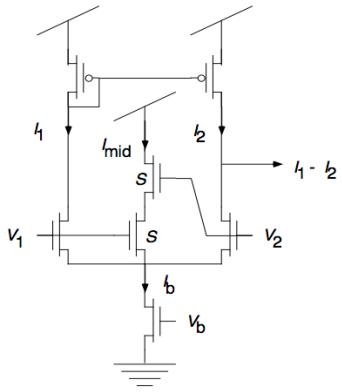
[5]

[bookwork]

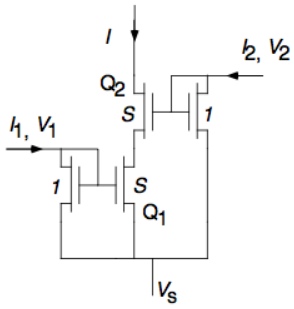
d) The system in Figure 6.1 uses a V/I converter with improved linearity over a conventional differential pair. Sketch suitable schematics for the V/I converter and the current correlator used in this system. [5]

[Application of Known Theory]

V/I = Bump transconductor[3]



Current correlator = Two transistor correlator [2]



Solutions